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Multimodal Generalized Defect Category Discovery in industrial scenarios via defect-aware representation guided calibrated clustering

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ABSTRACT

Multimodal industrial surface inspection requires the simultaneous classification of normal samples and known defects, while discovering novel defects from 2D images and 3D point clouds. Existing methods typically fine-tune modality-specific pre-trained models under the supervision of auxiliary defect samples. However, the applicability of these methods is limited in defect-scarce multimodal industrial inspection, where supervision from limited defects is unreliable and per-modality fine-tuning for multimodal adaptation incurs high costs. In this study, we propose **Multimodal Generalized Defect Category Discovery (M-GDCD)**, a modality-adaptive paradigm that learns exclusively from normal samples and limited labeled defects to categorize known defects and reveal novel categories. M-GDCD first develops a defect-aware representation module trained solely on normal data to extract high-defect-probability features, which are then projected onto a unit hypersphere, thereby mitigating scale discrepancies across modalities without additional fine-tuning. Subsequently, M-GDCD employs calibrated clustering, using the scarce labeled defects as soft constraints instead of hard cluster centers to avoid unreliable supervision. When both modalities are available, the decision boundaries could be refined by margin-based calibration guided by cross-modal consistency to further improve performance. Finally, test samples are evaluated by measuring their distances to the multimodal defect-representation cluster centers. Extensive experiments on multimodal datasets including MVTec 3D-AD, Eyecandies, and MVTec AD, under two data-split protocols and three modality-combination settings, demonstrate that the proposed M-GDCD framework achieves superior classification performance across both multimodal 2D-3D and unimodal 2D detection scenarios.

1. Introduction

Industrial surface defect detection constitutes a critical part of quality control in manufacturing, aiming to accurately identify surface defects during production [1,2]. Since surface defects can manifest as a combination of appearance anomalies and geometric irregularities, recent research advocates for the multimodal utilization of 2D visible images (IMG) and 3D point clouds (PC) to exploit their complementary information for the effective detection of these diverse defects [3,4]. In addition to sensing modalities, industrial production lines often operate under open-set inspection conditions, where normal samples, known defects, and previously unseen defects exist simultaneously. Therefore, considering both sensing modalities and category-level inspection requirements, it is necessary to develop a multimodal open-set defect detection paradigm.

As shown in Fig. 1, existing multimodal defect detection approaches mainly focus on either identifying unknown anomalies [5–9] or classifying fine-grained defects [10–12], leveraging 2D images, 3D point clouds, or their combination. However, multimodal open-set defect detection, which requires the accurate classification of both known and previously unseen defect categories across single or coexisting modalities, has received limited research attention. A similar requirement has been explored in the context of natural scene classification under the Generalized Category Discovery (GCD) setting, which formulates a basic “representation + classification” paradigm with hybrid supervision to jointly perform *known-class classification* and *novel-class discovery*. Nevertheless, despite the conceptual relevance of this setting, applying the GCD paradigm to multimodal defect detection remains challenging. This challenge primarily stems from the nature of industrial defect data, with such data inherently involving multimodal forms and extreme

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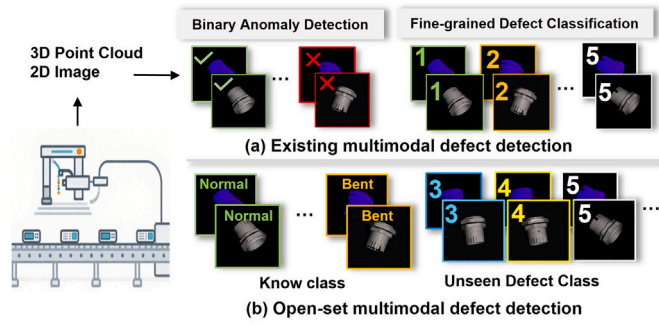


Fig. 1. Comparison of (a) Existing multimodal defect detection and (b) Open-set multimodal defect detection.

category imbalance. Specifically, these data are typically characterized by a dominance of normal samples, a severe scarcity of known defects, and the presence of certain defect categories that remain unobserved across single or coexisting modalities [3,13].

Specifically, industrial defect data pose two challenges for both representation and classification under this basic paradigm: (i) Multimodal representation gap between natural objects and industrial defects without additional priors: In natural scenes, most existing GCD methods [14–18] rely on fine-tuning pre-trained encoders with general semantic priors, assuming aligned priors between the encoder and natural images (Fig. 2(a)). In industrial scenarios, AnomalyNCD [19] also employs a GCD-style framework to accomplish novel class discovery, providing effective defect representations by leveraging its encoder with additional defect priors (Fig. 2(b)). However, these methods rely on large amounts of external data to provide priors, making fine-tuning challenging when known defects are limited. Moreover, since these methods primarily focus on image encoders, applying them to point cloud encoders demands specialized fine-tuning strategies, which leads to substantially higher annotation and training costs. (ii) Unstable multimodal defect classification under limited supervision: For classification, prior works usually use semi-supervised non-parametric clustering [14,20], which is limited in industrial scenarios because the limited known defects provide insufficient supervision for reliable classification [17]. Moreover, these classifiers are typically designed without multimodal awareness, limiting their ability to fully exploit diverse representations to improve performance.

To address the above challenges, we introduce **Multimodal Generalized Defect Category Discovery (M-GDCD)**, a modality-adaptive framework that learns from normal samples and limited labeled defects to identify known defects and discover novel categories. As shown in Fig. 2(c), M-GDCD leverages multimodal defect-aware representations combined with calibrated clustering. First, a multimodal defect-aware representation is constructed without relying on additional defect priors. Instead, high-likelihood defect regions are identified from the generated multimodal anomaly masks and their scores, and features from these regions are projected onto a hypersphere to form representations that generalize across modalities. Subsequently, these representations can be used within the multimodal calibrated clustering module to recognize known categories and discover novel ones. A few labeled defects can optionally serve as constraints for unimodal calibrated clustering to refine the distinction between known and novel categories. Notably, if all defects are assigned as new categories, classification can proceed directly without these labeled defect samples. When both image and point cloud modalities are available, unimodal calibration is extended to multimodal calibration, exploiting cross-modal consistency to further improve accuracy. Importantly, the entire pipeline operates on either one or both modalities without requiring large amounts of defect data

or additional training. In summary, the contributions of M-GDCD are as follows:

(1) To address the demands of open-set defect detection in industrial settings, we propose the Multimodal Generalized Defect Category Discovery (M-GDCD) framework. This modality-adaptive framework is designed to jointly identify normal samples, known defects, and novel categories by leveraging multimodal defect-aware representations guided by calibrated clustering.

(2) To establish a unified multimodal defect representation without relying on the priors of additional defect, M-GDCD introduces a defect-aware representation module trained solely on normal samples to estimate defect probabilities. For multimodal adaptation, high-defect-probability features from any modality are uniformly embedded in hyperspherical space without extra modality alignment or model retraining.

(3) To mitigate the unreliable supervision arising from limited known defects, M-GDCD adopts calibrated clustering that uses the limited defects as weak constraints rather than hard supervision typical of previous methods. By exploiting cross-modal consistency when multimodal data are simultaneously available, it further introduces margin-based calibration to refine decision boundaries and improve classification performance.

(4) Extensive experiments on diverse public benchmarks validate the superior performance of M-GDCD. Specifically, on the MVTEC 3D-AD benchmark, M-GDCD achieves state-of-the-art performance under two data splits and three modality combinations, with the most significant gains observed when multimodal data are fully available. Furthermore, additional evaluations on Eyecandies and MVTEC AD demonstrate that the proposed M-GDCD framework achieves applicability across both multimodal 2D-3D and unimodal 2D detection scenarios.

The rest of this paper is organized as follows: Section 2 provides a review of related works. Section 3 presents a detailed description of the proposed M-GDCD method. Section 4 demonstrates experimental results on public benchmarks, and Section 5 concludes the paper and discusses potential future research directions.

2. Related work

In this section, we review four research domains following coarse-to-fine classification and increasing modality. Specifically, Sections 2.1 and 2.2 review industrial anomaly detection methods using unimodal images and multimodal data, respectively. Section 2.3 focuses on fine-grained classification methods tailored for industrial defects. Importantly, Section 2.4 introduces the development of generalized category discovery, along with its applicability and challenges in industrial scenarios.

2.1. Industrial anomaly detection

Industrial anomaly detection aims to learn a unified normal representation that is significantly different from anomalies. These methods are primarily divided into two categories: embedding-based and reconstruction-based methods. Embedding-based methods leverage pre-trained encoders to distill normal and anomalous representations via feature retrieval [8], knowledge distillation [21], or distribution mapping [22]. Reconstruction-based methods reconstruct abnormal image regions or pre-trained features into a normal state via inpainting [23], deep feature reconstruction [24], and diffusion models [25], thereby generating significant reconstruction errors during inference. While the above methods remain predominantly image-based, an increasing number of multimodal studies are now exploiting complementary information from point clouds, text, or other modalities to further improve performance.

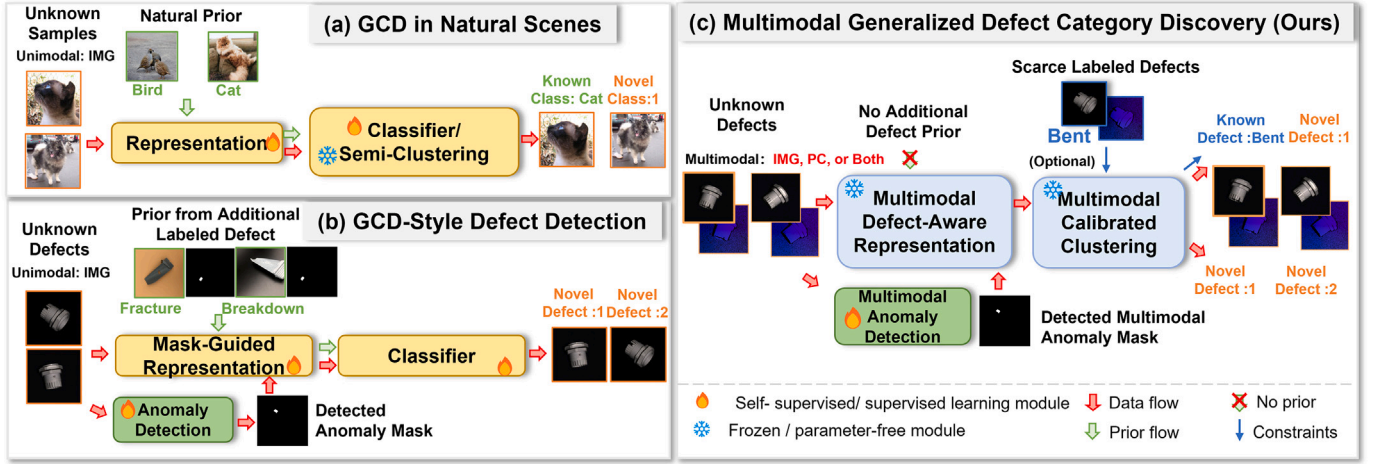


Fig. 2. Comparison of (a) GCD in natural scenes, (b) GCD-style defect detection, and (c) our multimodal generalized defect category discovery (M-DGCD).

2.2. Multimodal industrial anomaly detection

Multimodal detection has been widely explored in natural scene understanding, leveraging thermal cues in RGB-T settings [26–28] and geometric information in RGB-D scenarios [29]. These approaches, by combining semantic and structural representations through cross-modal and cross-scale fusion, consistently outperform single-modality methods under challenging conditions [30–32]. Benefiting from these advances, industrial anomaly detection has increasingly adopted multimodal strategies, integrating complementary sensor information and knowledge transferred from large-scale natural data to improve defect discrimination and generalization [4,33,34].

In industrial scenarios, multimodal data usually take the form of images and point clouds. A significant amount of research has focused on how to construct an accurate fusion representation to enhance the accuracy and robustness of models [35,36]. In terms of data-level fusion, AST [6] concatenated point clouds and images directly or after mapping, and then processed them in the same way as image-based anomaly detection. Additionally, some studies [37,38] focused on feature-level fusion to learn accurate multimodal representations for joint anomaly detection. Multimodal knowledge from natural scenes is mainly used to achieve highly generalized anomaly detection through vision-language models such as CLIP [39]. The unadjusted CLIP demonstrated certain anomaly detection capabilities [33,34]. Overall, the integration of both unique and general multimodal information contributes to the robustness and generalization of models, showing great potential in addressing complex scenarios. In addition, fine-grained defect classification of anomalies has been extensively studied to satisfy industrial defect inspection requirements.

2.3. Fine-grained industrial defect classification

This section reviews unsupervised learning methods for fine-grained classification of defects, which differ from supervised object detection models trained on labeled defect images. The key to defect classification lies in defining accurate representations and metrics. Early local representations focused on manually designed ones, such as three-dimensional textons used for representing and recognizing the visual appearance of materials [40]. AnomalyCluster [9] used a weighted Hausdorff distance to compute distances between patches for clustering, outperforming traditional Euclidean distances. Recently, unsupervised learning-based classification methods have proliferated. IIC [41] first unified representation learning and clustering in an end-to-end mutual-information objective. To relax this rigid constraint, GATCluster [42] replaced the direct optimization with attention-guided pseudo-labels. SCAN [43] established a paradigm for novel category discovery

by decoupling the pipeline, where a self-supervised backbone first acquired transferable features and a nearest-neighbor pseudo-labeling step subsequently adapts them to unseen classes.

However, these methods were primarily tailored for high-semantic natural image classification and performed poorly on low-semantic defect classification [9]. Meanwhile, Novel Class Discovery (NCD) was adopted in defect classification, transferring knowledge learned from labeled defects to classify unlabeled ones accurately [19]. While fine-grained industrial defect classification focuses on categorizing defects, it often overlooks the complexities of open industrial scenarios comprising normal samples, known defects, and unknown defects.

2.4. Generalized Category Discovery

Generalized Category Discovery (GCD) was first proposed under the assumption that unlabeled data come from both old and novel classes to address pragmatic classification in natural open-set scenarios [14]. Thus, the GCD formulation aligns with industrial scenarios where both known and unknown defects need to be detected. In natural scenarios, early research [14,20] focused on mining latent relationships to perform mix-supervised contrastive representation learning, followed by a non-parametric classifier. Subsequent works [15] shifted to parametric classifiers that separately assigned pseudo-labels to old and new classes. Also, a hybrid model [18] was proposed to integrate the above two strategies to further enhance performance. In summary, GCD methods employ a mix-supervised learning framework using both labeled and unlabeled data for representation and classification. However, in industrial scenarios, labeled defect data are extremely scarce, making it impractical to train a mix-supervised representation model and classifier. As industrial data extend beyond image modalities, training costs escalate rapidly with each additional modality. Based on this foundation, we further introduce Multimodal Generalized Defect Category Discovery (M-GDCD), a new task that demands a single framework to simultaneously confirm normal samples, re-identify previously seen defects, and discover previously unseen defects. To address challenges in representation and classification in M-GDCD, we propose defect-aware representation guided calibrated clustering that can effectively leverage the limited labeled defect samples and generalize across diverse data modalities.

3. Multimodal generalized defect category discovery

3.1. Problem formulation

3.1.1. Original setting of generalized category discovery

The original setting of Generalized Category Discovery (GCD) is presented with a partially-labeled dataset $\mathcal{D} = \mathcal{D}_l \cup \mathcal{D}_u$, where $\mathcal{D}_l =$

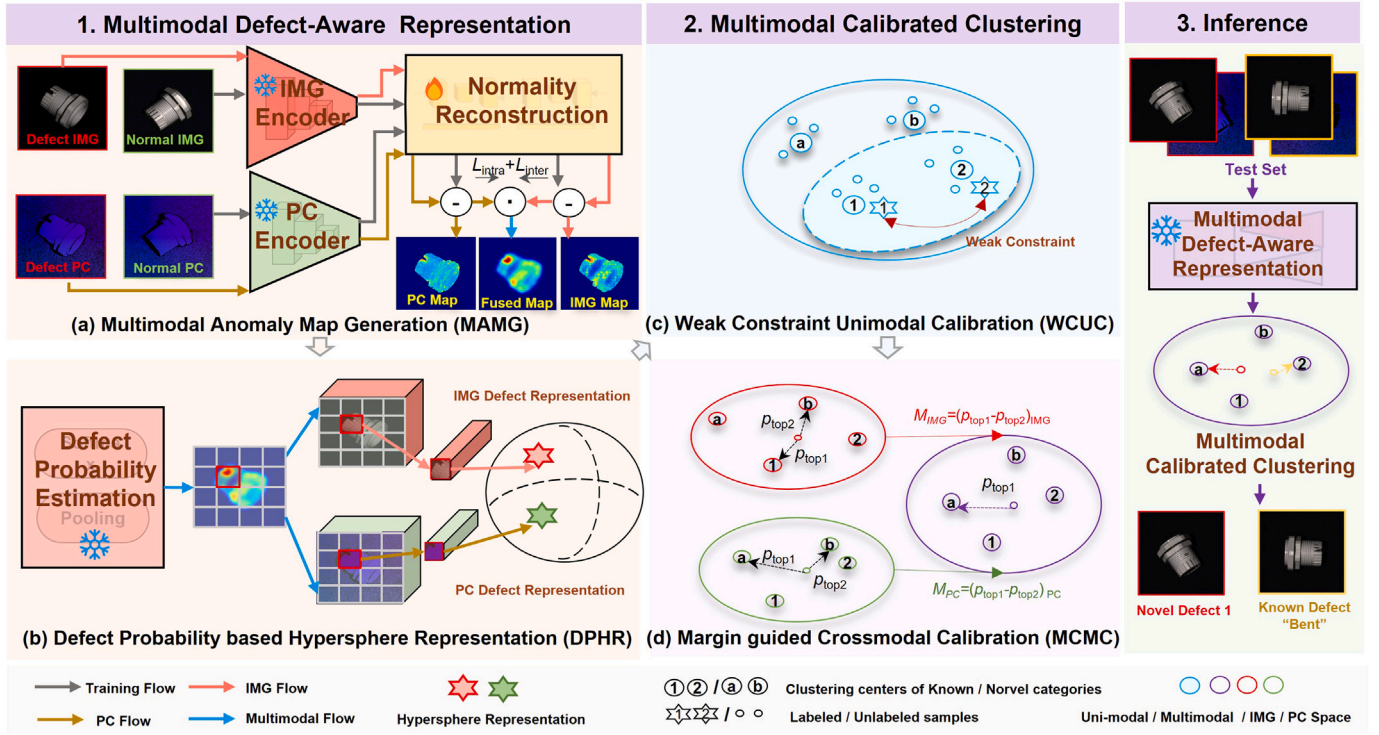


Fig. 3. M-GDCD Framework.

$\{(\mathbf{x}_i, y_i)\}_{i=1}^{N_l}$ and $D_u = \{\mathbf{x}_j\}_{j=1}^{N_u}$. Here, $\mathbf{x}_i$ denotes a labeled sample with label $y_i \in C_{old}$, and $\mathbf{x}_j$ denotes an unlabeled sample whose unknown label lies in $C_{old} \cup C_{new}$. The subscripts $i = 1, \dots, N_l$ and $j = 1, \dots, N_u$ index the labeled and unlabeled samples, respectively, and y_i, y_j are their corresponding ground-truth class labels. The objective is to classify unlabeled sample $\mathbf{x}_j$ into either a known class C_{old} or a novel class C_{new} .

3.1.2. Specialized setting for generalized defect category discovery

Building on the original GCD formulation, the specialized setup tailored for M-GDCD is characterized by differences in both data distribution and modality. In industrial defect inspection, the “normal” class is typically assigned as a known old class, while only a few defective samples are available. Thus, in the labeled set $D_l = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_l}$, we assign $y_i = 0$ to the *normal* class, which is present in abundance, whereas each known defect class contains only a few samples. Unlike the image-only modality commonly used in GCD, each industrial sample is represented by a multimodal observation $\mathbf{x} = (\mathbf{x}^{img}, \mathbf{x}^{pc})$, where $\mathbf{x}^{img} \in \mathbb{R}^{H \times W \times 3}$ is an RGB image (IMG) and $\mathbf{x}^{pc} \in \mathbb{R}^{N \times 3}$ is the corresponding point cloud (PC). The objective is to build a model using D_l that classifies each $\mathbf{x} \in D_u$ into the normal class, a known defect class, or a previously unseen defect class.

3.2. The architecture of M-GDCD

To achieve generalized classification by accurately leveraging normal samples and extremely limited known defects, the proposed M-GDCD framework is designed as a paradigm for multimodal defect-aware representation and calibrated clustering. As illustrated in Fig. 3, multimodal defect representation employs pre-trained multimodal features to model normality and measures the deviation of defect samples from this normality to estimate potential-defect probability, thereby guiding the selection of defect features and discriminative feature space mapping. After obtaining potential defect representations for a test sample, a few labeled prototypes within any single modality are used as weak constraints to calibrate the original clustering. Furthermore, when

multimodal data are available, we propose cross-modal calibration to determine the adaptive weight of each modality for classification based on its intra-domain separability margin. In addition, the framework can perform defect representation and generalized classification with only a single modality, except for the cross-modal calibration module. The components of the M-GDCD framework are described in detail as follows.

3.3. Multimodal Defect Representation

Multimodal Defect Representation aims to localize candidate defects via their discrepancy from normality and obtain discriminative representations via probability estimation-based hypersphere mapping. The details are provided below.

3.3.1. Multimodal anomaly map generation (MAMG)

Initially, modality-specific self-supervised encoders are employed to distill normality: images are first resized to $H \times W$, and point clouds are down-sampled to N_p points and projected onto the image plane, yielding paired inputs $\mathcal{X} = \{\mathbf{x}^{img}, \mathbf{x}^{pc}\}$. As shown in Fig. 3(a), the multimodal inputs $\mathcal{X} = \{\mathbf{x}^{img} \in \mathbb{R}^{H \times W \times 3}, \mathbf{x}^{pc} \in \mathbb{R}^{N \times 3}\}$ are encoded as:

$$\mathcal{Z} = \{\mathbf{z}^{img} = f_{img}(\mathbf{x}^{img}), \mathbf{z}^{pc} = f_{pc}(\mathbf{x}^{pc})\}, \quad (1)$$

where f_{img} and f_{pc} are the pretrained self-supervised encoders.

Next, inspired by [5,37,38], a lightweight reconstruction network g_{θ}^{rec} and a cross-modal mapping network g_{ϕ}^{trans} are jointly trained solely on normal embeddings $\{\mathbf{z}^{img}\}_{norm}$ and $\{\mathbf{z}^{pc}\}_{norm}$. These two networks are jointly trained via cosine similarity losses:

$$\begin{aligned} \mathcal{L}_{intra} &= \underbrace{\left[1 - \langle \mathbf{z}^{img}, g_{\theta}^{rec}(\mathbf{z}^{img}) \rangle\right]}_{img \text{ only}} + \underbrace{\left[1 - \langle \mathbf{z}^{pc}, g_{\theta}^{rec}(\mathbf{z}^{pc}) \rangle\right]}_{pc \text{ only}}, \\ \mathcal{L}_{inter} &= \underbrace{\left[1 - \langle \mathbf{z}^{pc}, g_{\phi}^{trans}(\mathbf{z}^{img}) \rangle\right]}_{img \rightarrow pc} + \underbrace{\left[1 - \langle \mathbf{z}^{img}, g_{\phi}^{trans}(\mathbf{z}^{pc}) \rangle\right]}_{pc \rightarrow img}. \end{aligned} \quad (2)$$

where $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$. The above losses are per-pixel and are averaged over the spatial scale $H \times W$ and batch size during training.

After that, we calculate the defect likelihood by aggregating the reconstruction errors within individual modalities and across modalities. Specifically, for each test sample, the embeddings are contrasted against the learned normality via intra- and cross-modal residuals as

$$\begin{aligned} \Delta^{\text{img}} &= \left\| g_\theta^{\text{rec}}(\mathbf{z}^{\text{img}}) - \mathbf{z}^{\text{img}} \right\|_2, & \Delta^{\text{pc}} &= \left\| g_\theta^{\text{rec}}(\mathbf{z}^{\text{pc}}) - \mathbf{z}^{\text{pc}} \right\|_2, \\ \Delta^{\text{img} \rightarrow \text{pc}} &= \left\| g_\phi^{\text{trans}}(\mathbf{z}^{\text{img}}) - \mathbf{z}^{\text{pc}} \right\|_2, & \Delta^{\text{pc} \rightarrow \text{img}} &= \left\| g_\phi^{\text{trans}}(\mathbf{z}^{\text{pc}}) - \mathbf{z}^{\text{img}} \right\|_2. \end{aligned} \quad (3)$$

Finally, the anomaly map is obtained as

$$\Delta = \begin{cases} \Delta^{\text{img}}, & \text{img only,} \\ \Delta^{\text{pc}}, & \text{pc only,} \\ (\Delta^{\text{img}} + \Delta^{\text{pc} \rightarrow \text{img}})(\Delta^{\text{pc}} + \Delta^{\text{img} \rightarrow \text{pc}}), & \text{both.} \end{cases} \quad (4)$$

3.3.2. Defect probability based hypersphere representation (DPHR)

First, the defect probability is assessed to pinpoint the locations and relative contributions of the underlying features. Following the Main Element Binarization method [19], we obtain the stable defect region by retaining only those residuals Δ that fall within the searched range $\beta = [\beta_{\text{low}}, \beta_{\text{high}}]$. Therefore, the defect probability is finally estimated as:

$$p_{\text{defect}} = \frac{1}{1 + \exp(-\tau_1 \Delta)}, \quad (5)$$

where τ_1 is a temperature parameter that controls the sharpness or flatness of the probability distribution.

Second, to retain the most defect-informative features, we first interpret the pooling operators in a patch-wise manner. For images, appearance and color features exhibit homogeneous spatial saliency. Therefore, we apply patch-level adaptive average pooling $\mathcal{A}_{\text{mean}}(\cdot)$ over a $p \times p$ window, which aggregates these features into a robust semantic descriptor of anomalous regions. Conversely, point clouds are dominated by sparse and highly discriminative geometric features. Thus, we employ patch-level max pooling $\mathcal{A}_{\text{max}}(\cdot)$ to selectively amplify salient structural features while suppressing low-informativity points.

Finally, the hypersphere representations of images and point clouds are obtained to further eliminate modality-specific scale differences. Formally, these representations are obtained by ℓ_2 -normalizing the defect-probability-weighted average features:

$$\mathbf{r}^{\text{img}} = \frac{\mathcal{A}_{\text{mean}}(\mathbf{z}^{\text{img}} \odot p_{\text{defect}}^{\text{img}})}{\|\mathcal{A}_{\text{mean}}(\mathbf{z}^{\text{img}} \odot p_{\text{defect}}^{\text{img}})\|_2}, \quad \mathbf{r}^{\text{pc}} = \frac{\mathcal{A}_{\text{max}}(\mathbf{z}^{\text{pc}} \odot p_{\text{defect}}^{\text{pc}})}{\|\mathcal{A}_{\text{max}}(\mathbf{z}^{\text{pc}} \odot p_{\text{defect}}^{\text{pc}})\|_2}, \quad (6)$$

where

$$\begin{aligned} \mathcal{A}_{\text{mean}}(\cdot) &= \frac{1}{p^2} \sum_{h=i}^{(i+1)p-1} \sum_{w=j}^{(j+1)p-1} (\cdot)_{h,w,c}, \\ \mathcal{A}_{\text{max}}(\cdot) &= \max_{\substack{h=i \\ w=j}}^{(i+1)p-1, (j+1)p-1} (\cdot)_{h,w,c}. \end{aligned} \quad (7)$$

3.4. Multimodal calibrated clustering

As shown in Fig. 3(c) and (d), we introduce multimodal calibrated clustering for the classification of known and unknown defects. Instead of using scarce known samples as cluster centroids, we use them as weak constraints to reduce bias. This approach can be easily applied when any single modality is available without further adjustments. When multiple modalities coexist, we further propose a margin-based cross-modal calibration that refines the fusion representations to classify defects more accurately.

3.4.1. Weak constraint-guided uni-modal calibration (WCUC)

To prevent unreliable supervision from dominating the clustering, we use labeled samples as *weak constraints* rather than hard centroids. Let C_{old} denote the set of old classes with available labels. For each class $k \in C_{\text{old}}$, the mean representation of labeled samples is computed as:

$$\mu_k^0 = \frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{r}_{k,i}, \quad (8)$$

where N_k is the number of labeled samples in class k , and $\mathbf{r}_{k,i}$ is the feature vector of the i th labeled sample.

Instead of enforcing μ_k^0 as a fixed centroid, we initialize standard K-Means to obtain initial cluster centers $\{\mu_k\}$ and assignment indicators $\mathbb{I}_{ik}$. Conflicting pairs (k_1, k_2) are identified whenever labeled samples from different classes are assigned to the same cluster.

For each cluster k , we define the within-cluster dispersion:

$$\delta_k = \frac{1}{|S_k|} \sum_{i \in S_k} \|\mathbf{r}_i - \mu_k\|_2, \quad (9)$$

where $S_k = \{i : \mathbb{I}_{ik} = 1\}$ is the set of points currently assigned to cluster k . This is simply the mean Euclidean distance of points to their centroid. For each conflicting pair (k_1, k_2) , we define a repulsion distance:

$$d^* = \gamma(\delta_{k_1} + \delta_{k_2}), \quad (10)$$

where $\gamma > 0$ is a hyperparameter controlling the strength of the repulsion. Whenever the current centroid distance $d = \|\mu_{k_2} - \mu_{k_1}\|_2 < d^*$, the centroids are pushed apart along the unit vector

$$\mathbf{v} = \frac{\mu_{k_2} - \mu_{k_1}}{\|\mu_{k_2} - \mu_{k_1}\|_2} \quad (11)$$

via:

$$\mu_{k_1} \leftarrow \mu_{k_1} - \frac{d^* - d}{2} \mathbf{v}, \quad \mu_{k_2} \leftarrow \mu_{k_2} + \frac{d^* - d}{2} \mathbf{v}. \quad (12)$$

This repulsion step is iteratively alternated with label reassignment. After each centroid update, samples are reassigned to their nearest cluster centers. The process repeats until convergence, which is defined as either (i) the maximum centroid displacement falling below a pre-defined tolerance $\varepsilon = 10^{-4}$, or (ii) the cluster assignments remaining unchanged between successive iterations.

3.4.2. Margin-guided cross-modal calibration (MCMC)

When two modalities coexist, cross-modal consistency can suppress domain shift and align heterogeneous feature spaces, thereby boosting classification performance. Therefore, we perform margin-guided cross-modal calibration on hypersphere representations.

First, for each sample i with paired multimodal representations $\mathbf{r}_i^{\text{img}}$ and $\mathbf{r}_i^{\text{pc}}$, we obtain the modality-specific cluster centers via the above WCUC. We then compute multimodal margins that quantify the modal confidence for every sample i :

$$M_i = \underbrace{\left(\max_c \mu_c^\top \mathbf{r}_i \right)}_{\text{top-1}} - \underbrace{\left(\max_{c \neq \arg \max_k \mu_k^\top \mathbf{r}_i} \mu_c^\top \mathbf{r}_i \right)}_{\text{top-2}} \quad (13)$$

Second, these margins M_i^{img} and M_i^{pc} reflect the confidence of each modality for sample i and are subsequently used to re-weight their contributions during fusion. We apply a temperature-scaled Softmax function to the margins to obtain the fusion weights:

$$w_i^{\text{img}}, w_i^{\text{pc}} = \text{softmax}([M_i^{\text{img}}/\tau_2, M_i^{\text{pc}}/\tau_2]), \quad (14)$$

where τ_2 is a temperature parameter that controls the flatness of the weight distribution.

Next, we leverage these margins to calibrate the final fused representation. Specifically, we project $\mathbf{r}^{\text{pc}}$ to the image feature space

Table 1
Dataset splits and modality configurations for the experiments.

Dataset	MVTec-3D AD	Eyecandies	MVTec AD
Modality	IMG+PC	IMG+PC	IMG
Object(Samples)	10(1197)	10(400)	15(1888)
CDC Split Per Object			
C_{old} (Normal)	1	1	1
C_{new} (Defects)	4–6	4	3–8
D_l	1	1	1
D_u	~100	25	~100
GDC Split Per Object			
C_{old}	2–3	2	2–4
C_{new}	2–3	3	2–4
D_l	2–3	2	1
D_u	~100	23	~100

through a hyperspherical semantic alignment network $h_{\theta}^{pc \rightarrow img}$, which is optimized via:

$$\mathcal{L}_{align} = \frac{1}{N} \sum_{i=1}^N (1 - \langle \mathbf{r}_i^{img}, h_{\theta}^{pc \rightarrow img}(\mathbf{r}_i^{pc}) \rangle). \quad (15)$$

The final calibrated fused representation for sample i is their weighted concatenation:

$$\mathbf{r}_i^{img+pc} = w_i^{img} \mathbf{r}_i^{img} \oplus w_i^{pc} h_{\theta}^{pc \rightarrow img}(\mathbf{r}_i^{pc}). \quad (16)$$

3.5. Training and inference

The proposed M-GDCD model exhibits flexible multimodal adaptability. When only single-modality data are available, only the unimodal reconstruction network $g_{\theta}^{rec}(\cdot)$ is activated and optimized using the unimodal loss in Eq. (2). If multimodal data are simultaneously accessible, the cross-modal mapping network $g_{\theta}^{trans}(\cdot)$ is additionally introduced while retaining the unimodal reconstruction module, fully exploiting complementary information across modalities. Concurrently, the defect-representation alignment network $h_{\theta}^{pc \rightarrow img}(\cdot)$ is optimized by minimizing Eq. (15). During inference, the fused multimodal representations are fed into the proposed calibrated clustering algorithm to iteratively achieve defect clustering.

4. Experiment

4.1. Dataset setup and evaluation metric

4.1.1. Dataset setup

We conduct experiments on the public MVTec-3D AD [4], Eyecandies [47], and MVTec [48] datasets under two distinct data splits and three modality configurations. The specific details of these splits and configurations are listed in Table 1. First, in the **complete defect classification (CDC) split** setting, consistent with [19], for each of the ten object categories, only the normal class is designated as the known class C_{old} , whereas all defective samples are classified as novel classes C_{new} . Second, following the generalized category discovery protocol [14], we introduce a **general defect classification (GDC) split**, in which the normal class together with half of the defective classes are labeled as old classes, with the remaining defect types assigned as new. To simulate label scarcity, merely *one labeled sample* is provided per old class, constituting the labeled set D_l ; the remaining normal samples and all defective samples form the unlabeled set D_u . Experiments are carried out under three modality configurations: (i) image only (IMG), (ii) point cloud only (PC), and (iii) the combination of both modalities (IMG+PC). Under the transductive setting, the model is trained on the union $D_l \cup D_u$ and evaluated exclusively on the unlabeled set D_u . In addition, the large number of normal samples in the training set is not explicitly listed, as this is a common practice for generating anomaly masks.

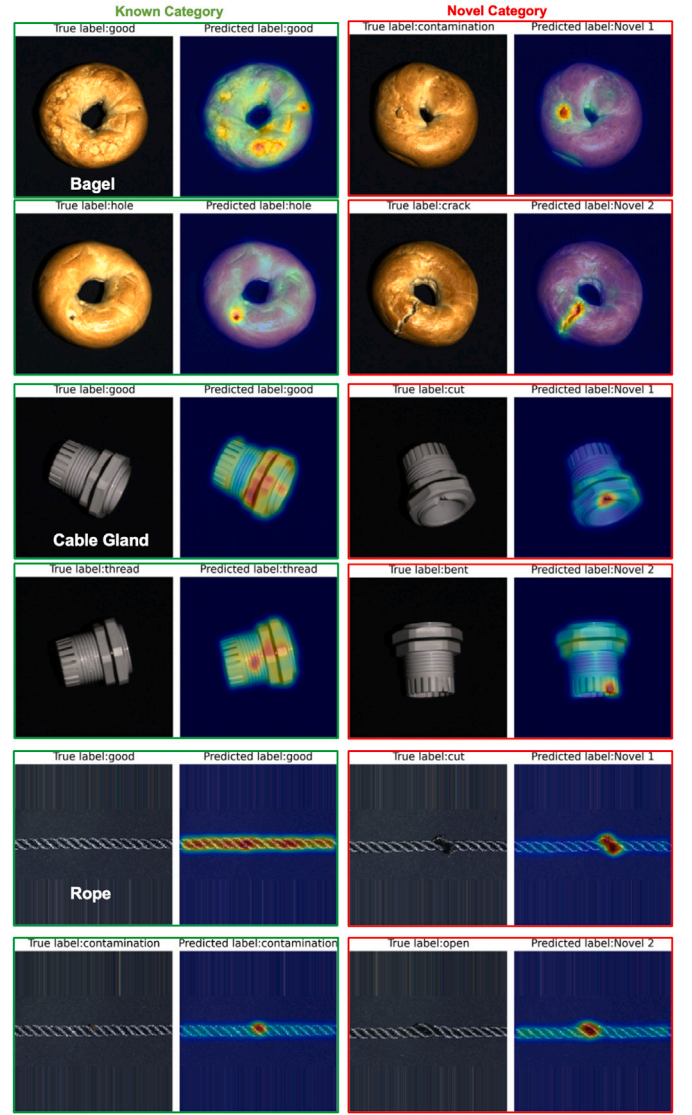


Fig. 4. Visualization results of our M-GDCD.

4.1.2. Evaluation metric

Following [9,19], we use the Hungarian algorithm to efficiently obtain the optimal one-to-one mapping between ground-truth and predicted labels. Performance is evaluated using three widely adopted metrics: the F1-score, Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI). Additionally, consistent with [14], we report the classification accuracies Acc_{old} , Acc_{new} , and Acc_{mean} for known (old) classes, novel (new) classes, and their average (mean), respectively. Notably, the CDC split uses the F1-score, NMI, and ARI to gauge how well the model distinguishes different fine-grained defect types, while the GDC split reports Acc_{old} , Acc_{new} , and Acc_{mean} to assess performance on both known and novel defect classes.

4.1.3. Implementation details

In this study, we employ Transformer-based backbones to extract both global and local features from industrial images and point clouds. Images are encoded by a DINO ViT-B/8 model pre-trained on ImageNet [49], while point clouds are encoded by a Point-MAE network pre-trained on ShapeNet [44]. During point cloud processing, background points lying within 0.005 units of the fitted plane are removed [37,38]. For network construction, the unimodal reconstruction

Table 2

Comparative results on MVTec 3D-AD under CDC split.

Modality	IMG					PC			IMG+PC			
Method	GCD [14]	SimGCD [15]	Anomaly Cluster [9]	Anomaly NCD [19]	M-GDCD (Ours)	PointMAE +K-means [44]	HDBSCAN [45]	M-GDCD (Ours)	MVSC [46]	GCD* [14]	Anomaly NCD*	M-GDCD (Ours)
F1	0.421	0.400	0.485	0.548	0.564	0.270	0.355	0.605	0.459	0.495	0.616	0.714
NMI	0.167	0.203	0.272	0.335	0.411	0.046	0.535	0.484	0.217	0.257	0.418	0.606
ARI	0.104	0.100	0.191	0.260	0.328	-0.001	0.305	0.396	0.157	0.170	0.349	0.532

Table 3

Comparative results on MVTec 3D-AD under GDC split with five random samplings.

Modality	IMG			PC		IMG+PC	
Method	GCD [14]	AnomalyNCD [19]	M-GDCD (Ours)	Point MAE+GCD [44]	M-GDCD(Ours)	AnomalyNCD* [19]	M-GDCD (Ours)
Acc _{old}	0.396±0.091	0.484±0.149	0.749±0.115	0.328 ±0.167	0.718±0.096	0.576±0.112	0.826±0.107
Acc _{new}	0.599±0.126	0.608±0.124	0.442±0.158	0.275 ±0.148	0.664±0.143	0.651±0.181	0.621±0.176
Acc _{mean}	0.501±0.079	0.545±0.102	0.590±0.086	0.303 ±0.247	0.584±0.086	0.612±0.116	0.717±0.126

network $g_{\theta}^{\text{rec}}(\cdot)$, the cross-modal reconstruction network $g_{\phi}^{\text{trans}}(\cdot)$, and the defect-representation alignment network $h_{\theta}^{\text{pc} \rightarrow \text{img}}(\cdot)$ are all implemented as three-layer MLPs with matching input and output dimensions. The number of channels in the hidden layers is set to half the sum of the input and output dimensions. All networks are trained for 250 epochs using the Adam optimizer with a learning rate of 0.001. The default settings for the temperature hyperparameters τ_1 and τ_2 are both set to 1, and γ is set to 2. Experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU.

4.1.4. Baseline settings

Since dedicated baselines for multimodal open-set industrial defect detection are limited, we adapted several representative methods from both the natural and industrial domains for comparison. For the image modality, we compared our method with general GCD-based methods, including GCD [14] and SimGCD [15], as well as industrial-specific approaches such as Anomaly Cluster [9] and Anomaly NCD [19]. Among them, GCD [14], SimGCD [15], and Anomaly NCD [19] require additional training on the AeBAD-S dataset [50], whereas our method does not rely on any extra defect data. Anomaly Cluster [9], which is fully unsupervised, was also included as a baseline commonly used in industrial scenarios. For the point cloud modality, we employed Point-MAE with K-means clustering [44] and the density-based clustering method HDBSCAN [45]. In the multimodal setting, we evaluated multi-view spectral clustering (MVSC) [46] alongside state-of-the-art methods adapted to multimodal inputs. Specifically, since GCD* [14] and Anomaly NCD* [19] do not natively support point cloud modality, we leveraged the multimodal anomaly maps provided by our method to generate corresponding region masks for inference. MVSC [46] uses features from different modalities as distinct views for joint clustering.

As a unified anomaly map is not provided by some baselines, methods that require precomputed anomaly maps used those generated by our MAMG module. The anomaly maps produced by different methods and their adaptation in our framework are further discussed in the extensibility experiments. All models followed their default training settings under CDC splits according to the protocols in Anomaly NCD, while under GDC splits, one labeled sample per old class was incorporated for classification to distinguish new defects from known ones.

4.2. Comparative experimental results

4.2.1. Classification performance comparison

As summarized in Table 2, compared with existing state-of-the-art (SOTA) methods, the proposed M-GDCD achieves consistent performance improvements across all modalities. Specifically, in the IMG modality, the F1, NMI, and ARI metrics of the proposed M-GDCD increase by 0.16, 0.76, and 0.68, respectively, over the second-place competitor, Anomaly NCD. Meanwhile, in the PC modality, M-GDCD

Table 4

Adaptability of M-GDCD to different multimodal anomaly detection methods on different datasets.

Dataset	MVTec-3D AD (IMG+PC)		MVTec-AD (IMG only)	
Method	M3DM [35] +Ours	CFM [37] +Ours	MuSc [51] +Ours	CPR [52] +Ours
F1	0.655	0.701	0.768	0.827
NMI	0.538	0.594	0.716	0.814
ARI	0.452	0.504	0.665	0.782
Acc _{mean}	0.674	0.696	0.759	0.840
Acc _{old}	0.741	0.750	0.806	0.867
Acc _{new}	0.612	0.644	0.717	0.812

Table 5

Performance on Eyecandies dataset.

Modality	F1	NMI	ARI	Acc _{mean}	Acc _{old}	Acc _{new}
IMG	0.535	0.616	0.785	0.539	0.722	0.417
PC	0.521	0.597	0.783	0.541	0.699	0.435
IMG+PC	0.537	0.626	0.786	0.548	0.717	0.436

achieves significant gains of 0.25 in F1 and 0.91 in ARI compared to HDBSCAN, although it is slightly lower in NMI. This is mainly because HDBSCAN is a density-based clustering method that provides a good distribution but tends to over-segment clusters. Furthermore, M-GDCD achieves the best performance in the IMG+PC modality, yielding specific values of 0.714, 0.606, and 0.532.

Table 3 summarizes the comparative performance on the MVTec3D-AD dataset under the GDC split. To mitigate potential bias stemming from specific labeled samples, all results are averaged over five random runs. For known defects, M-GDCD outperforms existing methods in Acc_{old} by 0.300. This is due to the defect-aware representations which reduce the impact of unreliable labels. Although the Acc_{new} of M-GDCD is slightly lower than that of the Anomaly NCD, M-GDCD achieves a better classification balance between old and new classes. Consequently, M-GDCD delivers the best Acc_{mean} across all modalities, outperforming the second-best competitor by 0.045 in IMG, 0.281 in PC, and 0.105 in IMG+PC. Some visualization results are shown in Fig. 4.

4.2.2. Generalization performance across modalities and datasets

To evaluate the generalization performance across various modalities and datasets, we report results on MVTec-3D AD, MVTec-AD, and Eyecandies. As summarized in Table 4, our method is integrated with M3DM [35] and CFM [37], achieving competitive performance compared with our original configurations on MVTec-3D AD. On MVTec-AD, when combined with state-of-the-art methods MuSc [51] and CPR [52], our approach yields strong performance in the unimodal (IMG-only) scenario. Table 5 reports results on Eyecandies, where each category contains only approximately five test defect samples, leading

Table 6
Complexity comparison.

Method	Additional dataset	Model size (MB)
GCD [14]	✓	3274.0
AnomalyCluster [9]	×	–
AnomalyNCD [19]	✓	4458.8
M-GDCD _{uni} (Ours)	×	431.8
M-GDCD _{mul} (Ours)	×	863.6

to slightly lower scores. Specifically, the F1, NMI, ARI, Acc_{mean} , Acc_{old} , and Acc_{new} of 0.521, 0.597, 0.783, 0.539, 0.699, 0.417 to 0.537, 0.626, 0.786, 0.548, 0.717, 0.436 are achieved, with the combined IMG+PC modalities providing the best overall results.

4.2.3. Model complexity comparison

As shown in Table 6, M-GDCD eliminates the need for additional defect datasets that are indispensable for auxiliary-training in both GCD and AnomalyNCD. Because these two competing methods must train an expert model for each of the ten product categories, they consume more than 3 GB of storage and incur heavy computational costs. Although AnomalyCluster is almost storage-free, it pays for this with lower accuracy. In contrast, our single-modal variant M-GDCD_{uni} keeps the representation model frozen and only adds a three-layer MLP, raising the overhead by roughly several MB. Even the multi-modal version M-GDCD_{mul} doubles the footprint than M-GDCD_{uni}, the total remains orders of magnitude smaller than the per-category model banks required by GCD and AnomalyNCD. These results confirm that the proposed M-GDCD requires less training time and model storage, which also demonstrates potential for deployment in resource-limited inspection scenarios.

Table 7 summarizes the computational requirements of M-GDCD across different modalities. Most of the computation arises from the backbone networks, while the three-layer MLP heads contribute a negligible overhead. For the single-modal image input, the ViT-B/8 backbone dominates the 66.9 GFLOPs, with a total of 85.8 million parameters and 349.2 MB of memory, processing 100 samples in 1.1 s at 40.8 FPS. The point cloud modality is lighter, with 6.0 GFLOPs and 22.0 million parameters, requiring 1.3 s per 100 samples at 37.9 FPS. Combining both modalities increases the FLOPs and parameters to 72.9 GFLOPs and 107.8 million, with 463.6 MB of memory and 3.5 s per 100 samples at 27.8 FPS, whereas the MLPs add only a small fraction of the total computation.

4.3. Ablation studies

Comprehensive ablation experiments were conducted to evaluate the effects of the four individual components of M-GDCD: multimodal anomaly map generation (MAMG), defect probability-based hypersphere representation (DPHR), weak constraint-guided unimodal calibration (WCUC), and margin-guided cross-modal calibration (MCMC). Furthermore, we provide a detailed analysis of the effectiveness of both the multimodal representation and the calibrated clustering.

4.3.1. Contribution of different components to overall performance gain

Table 8 quantifies the contribution of each M-GDCD component. As shown in cases A and B, relying solely on either MAMG or DPHR results in poor performance, because merely locating defect regions or performing hypersphere representation alone is insufficient to fully represent defects. In Case C, accurate defect representation is achieved through the fusion of MAMG and DPHR, delivering fundamental improvements across all metrics. Case D adds WCUC, raising the Acc_{mean} from 0.616 to 0.649, which confirms that weakly constrained clustering refines the known-novel boundary even when labels are extremely scarce. The variant without MCMC in Case E already achieves the best F1, NMI, and ARI among all variants. Case F then introduces MCMC,

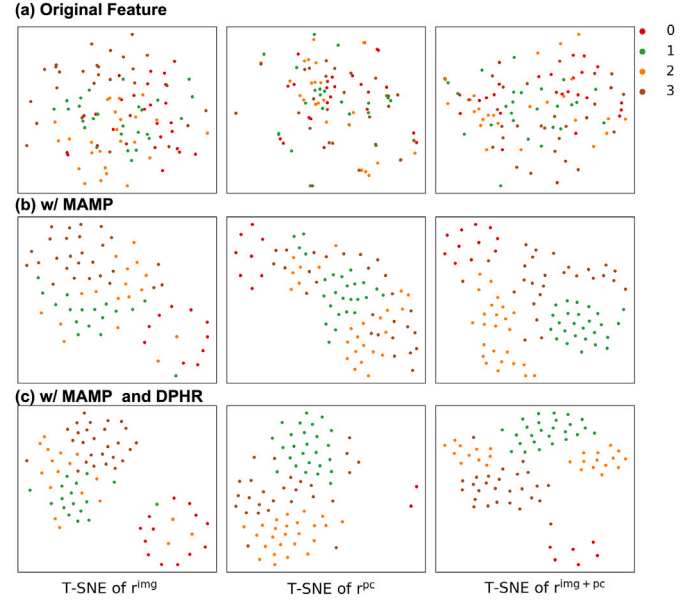


Fig. 5. T-SNE visualization of defect representations across modalities for the product category ‘cookie’.

improving Acc_{old} , Acc_{new} , and Acc_{mean} to 0.836, 0.632, and 0.730, respectively. In summary, each module contributes significantly, and the complete M-GDCD pipeline achieves the best overall performance.

4.3.2. Effectiveness of multimodal defect representations

To further analyze how the proposed modules improve the multimodal representations, we visualize various feature embeddings of the object category ‘cookie’ using t-SNE in Fig. 5. Original features are extracted directly from the image and point-cloud encoders, including their concatenation without any refinement. As shown in Fig. 5(a), due to the global semantic bias of the ViT-DINO [49], these vectors encode minimal low-level defect information, resulting in poorly separable feature clusters. In Fig. 5(b), we reweight the features using the anomaly maps from MAMG: background noise is suppressed, and activations focus on defect regions, greatly improving intra-class compactness and inter-class separability. To eliminate intensity variations caused by this weighting, we normalize the MAMG-refined features onto the unit hypersphere, yielding a clearly separated representation, as shown in Fig. 5(c). From a modal perspective, adding an extra modality consistently sharpens the anomaly maps and enhances the class-discriminative power of the embeddings at every stage.

Specifically, we evaluate the impact of different fusion strategies for the residual δ within the MAMP module. As shown in Table 9, our proposed fusion approach consistently achieves the best overall performance. Compared with the Additive (Add) and LogSumExp (LSE) strategies, our method demonstrates superior performance across all metrics.

Furthermore, we evaluated pooling strategies for each modality, as listed in Table 10. For single-modality input, the IMG modality is sensitive to the pooling strategy: using max-pooling (Max) instead of average-pooling (Avg) causes a notable drop in F1 from 0.561 to 0.523, and in Acc_{new} from 0.443 to 0.377, whereas the PC modality shows relatively minor fluctuations. For multimodal fusion, Max+Max and Avg+Max achieve similar performance, both significantly outperforming other combinations. Considering the requirement for balanced performance across modalities, we adopt IMG (Avg)+PC (Max) as the default setting for M-GDCD.

Finally, we evaluate different normalization strategies within the DPHR module. As listed in Table 11, the proposed ℓ_2 normalization

Table 7
Computational complexity for M-GDCD across different modalities.

Modality	Description	Representation				Clustering
		FLOPs	Params	Memory	FPS	
IMG	ViT-B/8+MLPs	66.9 G + 3.0 M	85.8 + 1.5 M	349.2MB	40.8	1.1s/100 samples
PC	PointMAE+MLPs	6.0 G + 4.4 M	22.0 + 2.2 M	96.8 MB	37.9	1.3s/100 samples
IMG+PC	Backbones(IMG+PC)+4*MLPs	72.9 G + 15.8 M	107.8 + 8.1M	463.6 MB	27.8	3.5s/100 samples

Table 8
Ablation experiments for different components of M-GDCD.

Module	Case					
	A	B	C	D	E	F
MAMG	✓	×	✓	✓	✓	✓
DPHR	×	✓	✓	✓	✓	✓
WCUC	×	×	×	✓	×	✓
MCMC*	×	×	×	×	✓	✓
F1	0.350	0.296	0.680	–	0.714	–
NMI	0.088	0.043	0.580	–	0.606	–
ARI	0.034	0.001	0.489	–	0.532	–
Acc _{old}	0.274	0.322	0.694	0.720	0.783	0.803
Acc _{new}	0.377	0.375	0.547	0.582	0.635	0.655
Acc _{mean}	0.322	0.350	0.616	0.649	0.706	0.725

“*” indicates that WCUC is not used, and a standard K-means/Semi-K-means classifier is adopted for the CDC/GDC splits. “–” denotes that WCUC is not evaluated, because only normal is known class in CDC setting (F1, NMI, ARI).

Table 9
Effect of different fusion strategies on the residual Δ in MAMP under IMG+PC modality.

Method	F1	NMI	ARI	Acc _{mean}	Acc _{old}	Acc _{new}
Add	0.597	0.475	0.394	0.612	0.651	0.566
LSE	0.658	0.511	0.455	0.663	0.761	0.572
Ours	0.711	0.611	0.529	0.724	0.805	0.651

Table 10
Comparison of different pooling strategies for each modality in DPHR.

Pooling	F1	NMI	ARI	Acc _{mean}	Acc _{old}	Acc _{new}
IMG						
Max	0.523	0.403	0.321	0.568	0.767	0.377
Avg	0.561	0.417	0.334	0.579	0.722	0.443
PC						
Max	0.582	0.477	0.386	0.601	0.774	0.446
Avg	0.585	0.466	0.380	0.610	0.748	0.490
IMG+PC						
Max+Max	0.723	0.615	0.535	0.731	0.818	0.650
Avg+Avg	0.677	0.578	0.487	0.687	0.772	0.606
Max+Avg	0.697	0.569	0.487	0.692	0.765	0.623
Avg+Max	0.721	0.615	0.537	0.724	0.805	0.651

consistently achieves the best overall performance. Compared to ℓ_1 normalization, ℓ_2 improves the F1, NMI, and ARI by 0.8%, 0.1%, and 0.6%, respectively, and yields a higher Acc_{mean} of 0.3% and Acc_{old} of 1.3%. While Z-score normalization exhibits comparable supervised performance, it is outperformed by ℓ_2 in terms of semi-supervised metrics, especially for Acc_{old}. In contrast, Min-Max normalization shows a clear performance degradation across all metrics. These results indicate that ℓ_2 normalization provides a more stable and discriminative feature scale for multimodal fusion within the DPHR module.

4.3.3. Effectiveness of calibrated clustering

Calibrated clustering consists of two main components: WCUC and MCMC. The former introduces weak constraints into the clustering process through a small set of labeled samples, whereas the latter adjusts the fusion weights of multimodal representations by exploiting the clustering margin. To investigate their contributions, we evaluate

Table 11
Effect of different feature normalization strategies in the DPHR.

Normalize	F1	NMI	ARI	Acc _{mean}	Acc _{old}	Acc _{new}
ℓ_1 Norm	0.612	0.498	0.409	0.637	0.759	0.524
Min-Max	0.589	0.491	0.398	0.615	0.729	0.512
Z-Score	0.611	0.498	0.414	0.637	0.759	0.523
ℓ_2 Norm(Ours)	0.620	0.499	0.415	0.640	0.772	0.520

Table 12
Comparison of different mapping strategies in MCMC.

Mapping	F1	NMI	ARI
IMG→PC	0.666	0.584	0.493
Concat	0.625	0.516	0.420
PC→IMG (Ours)	0.721	0.615	0.537

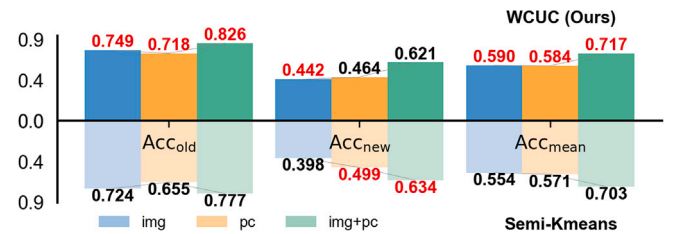


Fig. 6. Performance comparison between our WCUC and Semi-K-means under the same defect representations.

WCUC and the semi-K-means baseline from GCD under an identical multimodal defect representation. The key difference is that WCUC leverages labeled samples as clustering constraints, while semi-K-means only uses them to initialize cluster centroids. As shown in Fig. 6, when only one labeled sample per class is available, WCUC generally outperforms semi-K-means in terms of Acc_{mean}, Acc_{old}, and Acc_{new}. It shows pronounced improvements for the IMG and IMG+PC modalities, while maintaining comparable performance on PC.

Furthermore, we evaluated the effect of the repulsion hyperparameter γ on clustering performance across 10 random seeds by examining the mean accuracy comprising Acc_{mean}, Acc_{old}, and Acc_{new}, as well as their standard deviations, as shown in Fig. 7. Across all metrics, performance remains relatively stable for γ values from 0.1 to 2.0, with only slight fluctuations. Notably, when γ equals 2, Acc_{mean} reaches its maximum at 0.724 with a low standard deviation of 0.100. Larger values of γ , such as 5 and 10, result in a significant drop in performance, particularly in terms of the accuracy of new classes, suggesting over-smoothing or under-separation of clusters. Based on these observations, we select $\gamma = 2$ as the default setting. Across different random seeds and category distributions, the repulsion updates converge quickly, typically within five iterations, although the repulsion behavior may vary slightly across different seeds. Each iteration consumes less than 0.01 s per 100 samples, which is negligible relative to the overall computation.

To evaluate the effect of different feature mapping strategies in the MCMC module, we compare three alternatives: mapping image features to the point cloud space (IMG→PC), direct feature concatenation, and projecting point cloud features into the image space (PC→IMG). As shown in Table 12, the PC→IMG strategy achieves the best performance

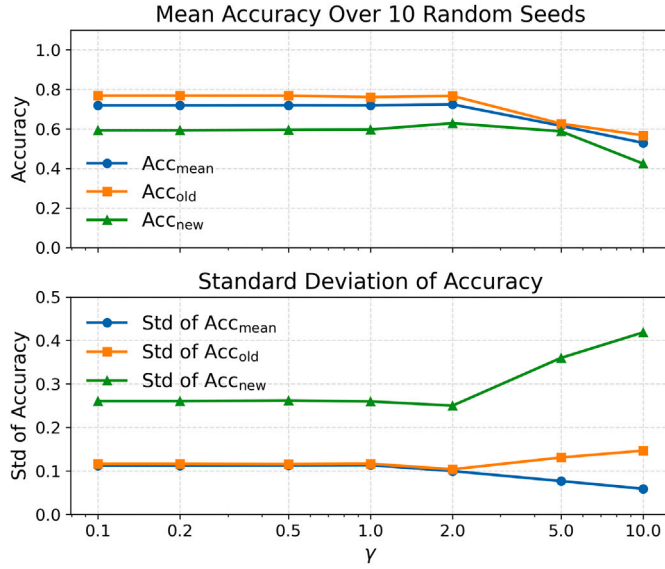
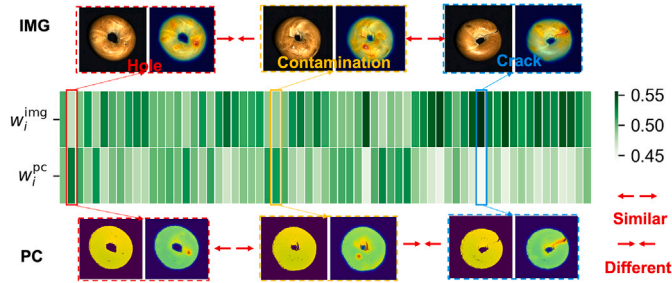
Fig. 7. Sensitivity Analysis of γ in the WCUC.

Fig. 8. Visualization of margin-based modality weights in MCMC.

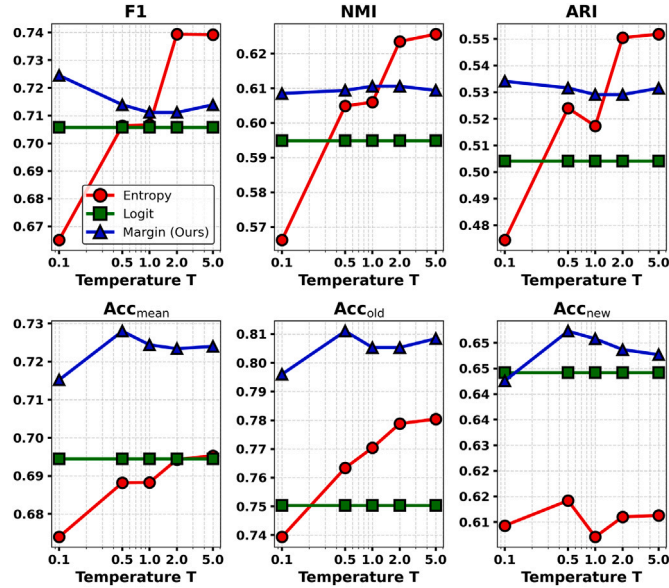


Fig. 9. Comparison of different fusion strategies in MCMC.

across all metrics, consistently outperforming the other approaches in terms of the F1, NMI, and ARI.

For the multimodal fusion in MCMC, Fig. 8 visualizes the margin-generated weights for all defect samples of the “bagel” category.

Table 13

Estimated number of clusters: MAE and relative deviation.

Method	MAE	Relative deviation
Silhouette	2.10	50%
GMM-BIC	2.00	48%
HDBSCAN	0.60	14%

Sample-wise heatmaps further reveal that hole and contamination defects exhibit similar visual patterns in images but distinct depth characteristics within point clouds. Conversely, as cracks are already highly discriminative in the 2D image modality, the corresponding point-cloud information is effectively redundant, consequently receiving minimal weight. These observations indicate that MCMC assigns lower weights to consistent modalities and higher weights to divergent ones, thereby calibrating the fused representation and boosting classification performance. We also evaluate the impact of different fusion strategies within the MCMC module under varying temperature T values, as shown in Fig. 9. Three strategies are compared, namely entropy-based, logit-based, and margin-based approaches. Specifically, entropy-based fusion shows a slightly higher F1, NMI, and ARI under the CDC setting at temperature values of 2 and 5, yet it is relatively sensitive to temperature variations. Logit-based fusion consistently exhibits lower performance across different temperatures, effectively reflecting an equal-weighted fusion behavior. Notably, margin-based fusion is relatively insensitive to temperature variations, consistently maintaining a balanced performance across both CDC and GDC settings, while achieving balanced performance between Acc_{old} and Acc_{new} under the GDC setting.

4.4. Effect of cluster number deviation on performance

To assess robustness to cluster-number variations, we examine how deviations between the assumed cluster number and the ground-truth category count impact clustering performance. As shown in Fig. 10, all modalities exhibit a consistent trend: performance metrics reach their peak when the cluster number matches the true number of categories, and gradually decrease as the deviation increases. This degradation mainly arises from incorrect cluster splitting or merging, which is further amplified by the limited sample size of certain categories. To enhance robustness in practical applications where the number of categories is unknown, we estimate the cluster number using the Silhouette Coefficient (Silhouette), the Gaussian Mixture Model Selection (GMM-BIC), and HDBSCAN. As reported in Table 13, HDBSCAN achieves the lowest mean absolute error (MAE) of 0.60 and a relative deviation of 14%, benefiting from a density-based mechanism that adapts to clusters of various shapes and sizes. This data-driven estimation provides valuable guidance for improving the robustness of our method.

4.5. Influence of modality

To further investigate how each modality contributes to overall performance, Fig. 11 quantifies the gains obtained by combining defect-probability estimation Δ in Eq. (4) and representation r in Eq. (6) across different modalities. Across all sub-plots the highest F1, NMI and ARI values consistently appear in the bottom-right cell, confirming that defect estimation and the final representation both achieve optimal performance when IMG+PC. Fixing the column and comparing rows across sub-figures shows performance under Δ^{img+pc} consistently exceeds Δ^{img} and Δ^{pc} , demonstrating that multimodal information yields more reliable defect probabilities. Fixing the row and comparing columns reveals the same consistent superiority for r^{img+pc} , indicating that multimodal fusion supplies richer and more separable representations. Therefore, these results confirm that joint multimodal defect estimation and representation consistently outperform any single-modal alternative, delivering both more reliable probabilities and more discriminative features.

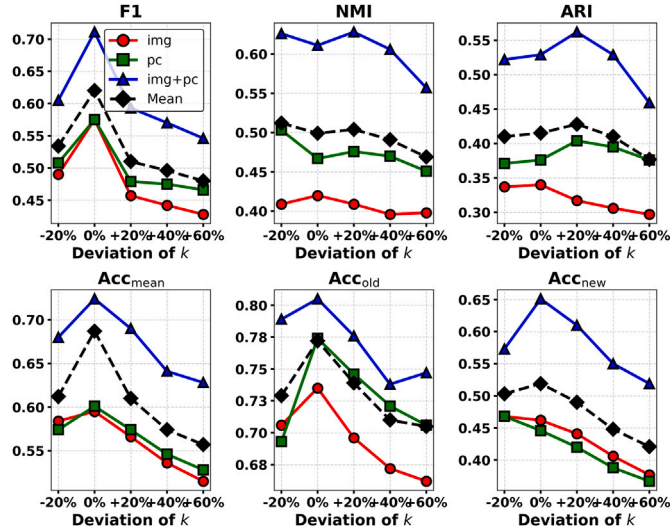


Fig. 10. Performance under Different Cluster Number Deviations.

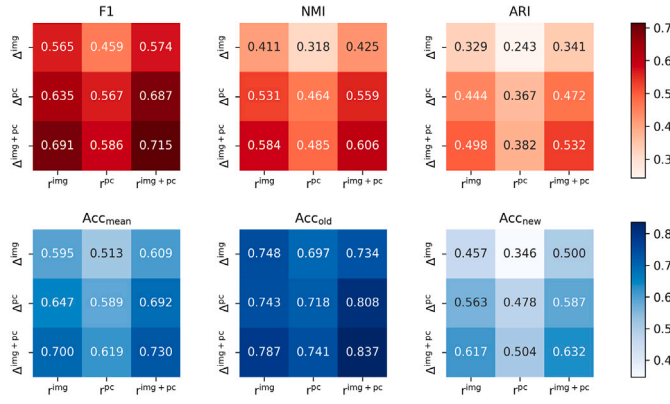


Fig. 11. Influence of modality.

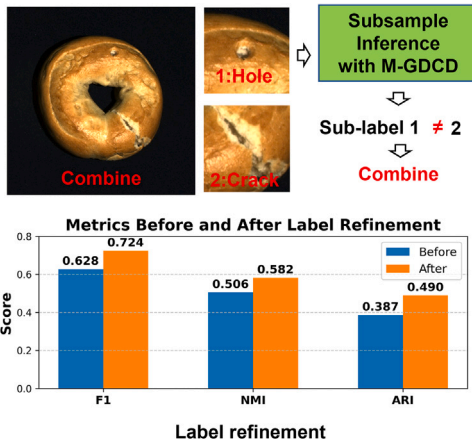


Fig. 12. Label refinement on combined defect detection.

4.6. Discussion on combined and missing defect detection

As illustrated in Fig. 12, our method performs label refinement to handle scenarios where multiple anomaly categories coexist within a single sample. Inspired by AnomalyNCD [19], features from each

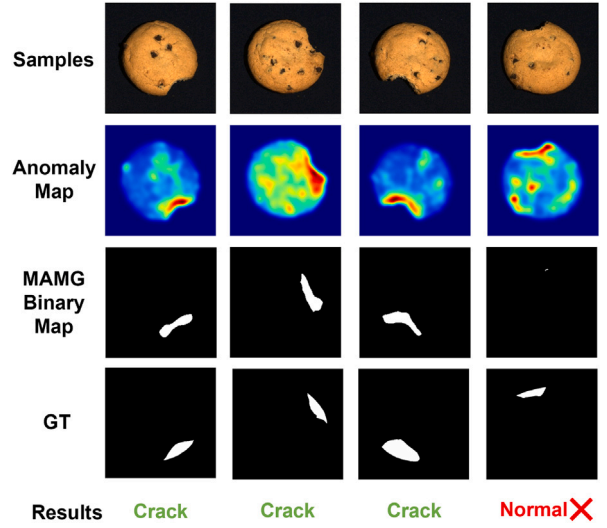


Fig. 13. Missing-type defects detection.

Table 14

Classification performance of the anomaly mask from the MAMG and ground-truth (GT).

Method	F1	NMI	ARI
MAMG	0.981	0.942	0.949
GT	0.991	0.971	0.975

connected region are labeled as individual subsamples, and hyperspherical representations are constructed for region-level classification. In the MVTec-3D AD dataset, a sample is categorized as a “combined” case if it contains two or more connected regions assigned to different subcategories. Corresponding experiments presented in Fig. 12 validate the effectiveness of this label refinement strategy.

In the “cookie” class of MVTec 3D-AD, the MAMG module produces anomaly masks for “missing” type defects that are sometimes only partially captured, as illustrated in Fig. 13. Despite this partial localization, the classification of these defects remains highly accurate. As shown in Table 14, clustering performance leveraging the MAMG masks is comparable to that obtained using ground-truth masks, achieving F1, NMI, and ARI scores of 0.981, 0.942, and 0.949, respectively, whereas the ground-truth masks yield scores of 0.991, 0.971, and 0.975. Specifically, only one sample is misclassified as normal because its binarization resulted in an excessively small partial mask, while all other instances are accurately classified.

4.7. Limitation

Although the proposed M-GDCD achieves strong performance across all modalities, its classification and generalization capabilities are still limited by the anomaly detection and binarization processes within the MAMG module. This limitation is particularly evident in irregular point cloud data, where accurate binarization is challenging due to the sparse and unstructured nature of the modality. In future work, we plan to investigate more robust methods for generalized defect category discovery, specifically tailored for sparse and unstructured point clouds, thereby extending the applicability of our framework to more challenging pure 3D defect detection tasks [53].

5. Conclusion

In this work, we introduce Multimodal Generalized Defect Category Discovery (M-GDCD), a modality-adaptive framework that simultaneously classifies normal samples, previously observed defects and

unforeseen defects without requiring external defect data or modality-specific fine-tuning. M-GDCD first employs a lightweight multimodal binary classifier from easily obtainable normal samples, and then derives defect-aware hypersphere representations. Extremely scarce labeled defects are employed as weak constraints inside a calibrated clustering module, which refines decision boundaries and extends to the multimodal adaptation through margin-based cross-modal consistency for further improve performance. The effectiveness of M-GDCD is validated through comprehensive evaluations on multimodal benchmarks such as MVTec 3D-AD, Eyecandies, and MVTec AD, where the framework exhibits superior classification performance across both multimodal 2D-3D and unimodal 2D detection tasks. In future work, we aim to develop a robust classification method for the more challenging field of pure 3D defect detection.

CRedit authorship contribution statement

Hao Cheng: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Jiaxiang Luo:** Writing – review & editing, Project administration, Funding acquisition. **Zilong Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Z. Zhang, Z. Zhao, X. Zhang, C. Sun, X. Chen, Industrial anomaly detection with domain shift: A real-world dataset and masked multi-scale reconstruction, *Comput. Ind.* 151 (2023) 103990, <http://dx.doi.org/10.1016/j.compind.2023.103990>.
- [2] H. Dong, K. Song, Y. He, J. Xu, Y. Yan, Q. Meng, PGA-Net: Pyramid feature fusion and global context attention network for automated surface defect detection, *IEEE Trans. Ind. Inf.* 16 (12) (2020) 7448–7458, <http://dx.doi.org/10.1109/TII.2019.2958826>.
- [3] M. Niu, K. Song, L. Huang, Q. Wang, Y. Yan, Q. Meng, Unsupervised saliency detection of rail surface defects using stereoscopic images, *IEEE Trans. Ind. Inf.* 17 (3) (2021) 2271–2281, <http://dx.doi.org/10.1109/TII.2020.3004397>.
- [4] P. Bergmann, X. Jin, D. Sattlegger, C. Steger, The mvtec 3D-AD dataset for unsupervised 3D anomaly detection and localization, in: *Proc. Int. Joint Conf. Comput. Vis. Imag. Comput. Graph. Theory Appl.*, 2022, pp. 202–213, <http://dx.doi.org/10.5220/0010865000003124>.
- [5] W. Shanguan, H. Wu, Y. Niu, H. Yin, J. Yu, B. Chen, B. Huang, CIPR: Multimodal industrial anomaly detection via latent bridged cross-modal prediction and intra-modal reconstruction, *Adv. Eng. Inf.* 65 (2025) 103240, <http://dx.doi.org/10.1016/j.aei.2025.103240>.
- [6] M. Rudolph, T. Wehrbein, B. Rosenhahn, B. Wandt, Asymmetric student-teacher networks for industrial anomaly detection, in: *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2023, pp. 2592–2602, <http://dx.doi.org/10.1109/WACV56688.2023.00262>.
- [7] Y. Cai, D. Liang, D. Luo, X. He, X. Yang, X. Bai, A discrepancy aware framework for robust anomaly detection, *IEEE Trans. Ind. Inf.* 20 (3) (2024) 3986–3995, <http://dx.doi.org/10.1109/TII.2023.3318302>.
- [8] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, P. Gehler, Towards total recall in industrial anomaly detection, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 14298–14308, <http://dx.doi.org/10.1109/CVPR52688.2022.01392>.
- [9] K. Sohn, J. Yoon, C.-L. Li, C.-Y. Lee, T. Pfister, Anomaly clustering: Grouping images into coherent clusters of anomaly types, in: *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2023, pp. 5468–5479, <http://dx.doi.org/10.1109/WACV56688.2023.00544>.
- [10] C. Pu, J. Wang, Y. Zhang, M. Niu, Q. Wu, Z. Lin, Geometric spatial constraints network for slender and tiny surface defect detection, *Adv. Eng. Inf.* 65 (2025) 103138, <http://dx.doi.org/10.1016/j.aei.2025.103138>.
- [11] J. Wang, K. Song, D. Zhang, M. Niu, Y. Yan, Collaborative learning attention network based on RGB image and depth image for surface defect inspection of no-service rail, *IEEE-ASME Trans. Mechatron.* 27 (6) (2022) 4874–4884, <http://dx.doi.org/10.1109/TMECH.2022.3167412>.
- [12] X. Ni, Z. Ma, J. Liu, B. Shi, H. Liu, Attention network for rail surface defect detection via consistency of intersection-over-union (IoU)-guided center-point estimation, *IEEE Trans. Ind. Inf.* 18 (3) (2021) 1694–1705, <http://dx.doi.org/10.1109/TII.2021.3085848>.
- [13] J. Luo, J. Lin, Z. Yang, H. Liu, SMD anomaly detection: A self-supervised texture-structure anomaly detection framework, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–11, <http://dx.doi.org/10.1109/TIM.2022.3194920>.
- [14] S. Vaze, K. Hant, A. Vedaldi, A. Zisserman, Generalized category discovery, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7482–7491, <http://dx.doi.org/10.1109/CVPR52688.2022.00734>.
- [15] X. Wen, B. Zhao, X. Qi, Parametric classification for generalized category discovery: A baseline study, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16590–16600, <http://dx.doi.org/10.1109/ICCV51070.2023.01521>.
- [16] Y. Liu, K. Han, DebGCD: Debaised learning with distribution guidance for generalized category discovery, in: *Proc. Int. Conf. Learn. Represent.*, 2025, pp. 16028–16041, <http://dx.doi.org/10.48550/arXiv.2504.04804>.
- [17] J. Bai, Z. Liu, H. Wang, R. Chen, L. Mu, X. Li, J.T. Zhou, Y. Feng, J. Wu, H. Hu, Towards distribution-agnostic generalized category discovery, *Adv. Neural Inf. Process. Syst.* 36 (2023) 58625–58647.
- [18] S. Ma, F. Zhu, X.-Y. Zhang, C.-L. Liu, ProtoGCD: Unified and unbiased prototype learning for generalized category discovery, *IEEE Trans. Pattern Anal. Mach. Intell.* 47 (7) (2025) 6022–6038, <http://dx.doi.org/10.1109/TPAMI.2025.3557502>.
- [19] Z. Huang, X. Li, H. Liu, F. Xue, Y. Wang, Y. Zhou, AnomalyNCD: Towards novel anomaly class discovery in industrial scenarios, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 4755–4765, <http://dx.doi.org/10.1109/CVPR52734.2025.00448>.
- [20] N. Pu, Z. Zhong, N. Sebe, Dynamic conceptional contrastive learning for generalized category discovery, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 7579–7588, <http://dx.doi.org/10.1109/CVPR52729.2023.00732>.
- [21] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, Uninformed students: Student-teacher anomaly detection with discriminative latent embeddings, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 4182–4191, <http://dx.doi.org/10.1109/CVPR42600.2020.00424>.
- [22] M. Rudolph, B. Wandt, B. Rosenhahn, Same same but DifferNet: Semi-supervised defect detection with normalizing flows, in: *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2021, pp. 1906–1915, <http://dx.doi.org/10.1109/WACV48630.2021.00195>.
- [23] V. Zavrtanik, M. Kristan, D. Škočaj, Reconstruction by inpainting for visual anomaly detection, *Pattern Recognit.* 112 (2021) 107706, <http://dx.doi.org/10.1016/j.patrec.2020.107706>.
- [24] Y. Shi, J. Yang, Z. Qi, Unsupervised anomaly segmentation via deep feature reconstruction, *Neurocomputing* 424 (2021) 9–22, <http://dx.doi.org/10.1016/j.neucom.2020.11.018>.
- [25] H. He, J. Zhang, H. Chen, X. Chen, Z. Li, X. Chen, Y. Wang, C. Wang, L. Xie, A diffusion-based framework for multi-class anomaly detection, in: *Proc. AAAI Conf. Artif. Intell.*, vol. 38, (8) 2024, pp. 8472–8480.
- [26] W. Zhou, Y. Zhu, J. Lei, R. Yang, L. Yu, LSNet: Lightweight spatial boosting network for detecting salient objects in RGB-thermal images, *IEEE Trans. Image Process.* 32 (2023) 1329–1340, <http://dx.doi.org/10.1109/TIP.2023.3242775>.
- [27] W. Zhou, F. Sun, Q. Jiang, R. Cong, J.-N. Hwang, WaveNet: Wavelet network with knowledge distillation for RGB-T salient object detection, *IEEE Trans. Image Process.* 32 (2023) 3027–3039, <http://dx.doi.org/10.1109/TIP.2023.3275538>.
- [28] W. Zhou, Q. Guo, J. Lei, L. Yu, J.-N. Hwang, ECFNet: Effective and consistent feature fusion network for RGB-T salient object detection, *IEEE Trans. Circuits Syst. Video Technol.* 32 (3) (2022) 1224–1235, <http://dx.doi.org/10.1109/TCSVT.2021.3077058>.
- [29] W. Zhou, Y. Zhu, J. Lei, J. Wan, L. Yu, CCAFFNet: Crossflow and cross-scale adaptive fusion network for detecting salient objects in RGB-D images, *IEEE Trans. Multimed.* 24 (2022) 2192–2204, <http://dx.doi.org/10.1109/TMM.2021.3077767>.
- [30] W. Zhou, Z. Ju, R. Cong, W. Yan, RCNet: Dual-network resonance collaboration via mutual learning for RGB-D road defect detection, *IEEE Trans. Circuits Syst. Video Technol.* (2025) 1, <http://dx.doi.org/10.1109/TCSVT.2025.3617769>.

- [31] W. Zhou, X. Yang, W. Yan, Q. Jiang, Hybrid knowledge distillation for RGB-T crowd density estimation in smart surveillance systems, *IEEE Internet Things J.* 12 (7) (2025) 9276–9289, <http://dx.doi.org/10.1109/JIOT.2024.3506624>.
- [32] W. Zhou, Y. Wang, X. Qian, Knowledge distillation and contrastive learning for detecting visible-infrared transmission lines using separated stagger registration network, *IEEE Trans. Circuits Syst. I. Regul. Pap.* 72 (8) (2025) 4140–4152, <http://dx.doi.org/10.1109/TCSI.2024.3521933>.
- [33] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, O. Dabeer, Winclip: Zero-/few-shot anomaly classification and segmentation, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 19606–19616, <http://dx.doi.org/10.1109/CVPR52729.2023.01878>.
- [34] Y. Cao, J. Zhang, L. Frittoli, Y. Cheng, W. Shen, G. Boracchi, Adaclip: Adapting clip with hybrid learnable prompts for zero-shot anomaly detection, in: *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 55–72.
- [35] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, C. Wang, Multimodal industrial anomaly detection via hybrid fusion, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8032–8041, <http://dx.doi.org/10.1109/CVPR52729.2023.00776>.
- [36] Y.-M. Chu, C. Liu, T.-I. Hsieh, H.-T. Chen, T.-L. Liu, Shape-guided dual-memory learning for 3D anomaly detection, in: *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 6185–6194.
- [37] A. Costanzino, P.Z. Ramirez, G. Lisanti, L. Di Stefano, Multimodal industrial anomaly detection by crossmodal feature mapping, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 17234–17243.
- [38] H. Cheng, J. Luo, X. Zhang, Multimodal industrial anomaly detection via uni-modal and cross-modal fusion, *IEEE Trans. Ind. Inf.* 21 (6) (2025) 5000–5010, <http://dx.doi.org/10.1109/TII.2025.3552723>.
- [39] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [40] T. Leung, J. Malik, Representing and recognizing the visual appearance of materials using three-dimensional textons, *Int. J. Comput. Vis.* 43 (1) (2001) 29–44, <http://dx.doi.org/10.1023/A:1011126920638>.
- [41] X. Ji, A. Vedaldi, J. Henriques, Invariant information clustering for unsupervised image classification and segmentation, in: *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 9864–9873, <http://dx.doi.org/10.1109/ICCV.2019.00996>.
- [42] C. Niu, J. Zhang, G. Wang, J. Liang, Gatcluster: Self-supervised gaussian-attention network for image clustering, in: *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 735–751.
- [43] W. Van Gansbeke, S. Vandenhende, S. Georgoulis, M. Proesmans, L. Van Gool, Scan: Learning to classify images without labels, in: *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 268–285.
- [44] Y. Pang, W. Wang, F.E. Tay, W. Liu, Y. Tian, L. Yuan, Masked autoencoders for point cloud self-supervised learning, in: *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 604–621.
- [45] R.J. Campello, D. Moulavi, A. Zimek, J. Sander, Hierarchical density estimates for data clustering, visualization, and outlier detection, *ACM Trans. Knowl. Discov. Data* 10 (1) (2015) 1–51, <http://dx.doi.org/10.1145/2733381>.
- [46] A. Kumar, P. Rai, H. Daume, Co-regularized multi-view spectral clustering, *Adv. Neural Inf. Process. Syst.* 24 (2011).
- [47] L. Bonfiglioli, M. Toschi, D. Silvestri, N. Fioraio, D. De Gregorio, The eyecandies dataset for unsupervised multimodal anomaly detection and localization, in: *Proceedings of the 16th Asian Conference on Computer Vision*, 2022, ACCV.
- [48] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, MVTec AD — A comprehensive real-world dataset for unsupervised anomaly detection, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9584–9592, <http://dx.doi.org/10.1109/CVPR.2019.00982>.
- [49] M. Caron, H. Touvron, I. Misra, H. Jegou, J. Mairal, P. Bojanowski, A. Joulin, Emerging properties in self-supervised vision transformers, in: *Proc. IEEE Int. Conf. Comput. Vis.*, 2021, pp. 9630–9640, <http://dx.doi.org/10.1109/ICCV48922.2021.00951>.
- [50] Z. Zhang, Z. Zhao, X. Zhang, C. Sun, X. Chen, Industrial anomaly detection with domain shift: A real-world dataset and masked multi-scale reconstruction, *Comput. Ind.* 151 (2023) 103990, <http://dx.doi.org/10.1016/j.compind.2023.103990>.
- [51] X. Li, Z. Huang, F. Xue, Y. Zhou, Musc: Zero-shot industrial anomaly classification and segmentation with mutual scoring of the unlabeled images, in: *The Twelfth International Conference on Learning Representations*, 2024.
- [52] H. Li, J. Hu, B. Li, H. Chen, Y. Zheng, C. Shen, Target before shooting: Accurate anomaly detection and localization under one millisecond via cascade patch retrieval, *IEEE Trans. Image Process.* 33 (2024) 5606–5621, <http://dx.doi.org/10.1109/TIP.2024.3448263>.
- [53] J. Liu, G. Xie, R. Chen, X. Li, J. Wang, Y. Liu, C. Wang, F. Zheng, Real3D-AD: A dataset of point cloud anomaly detection, in: A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, S. Levine (Eds.), *Adv. Neural Inf. Process. Syst.*, vol. 36, Curran Associates, Inc., 2023, pp. 30402–30415.